IDLAN ZAFRAN MOHD ZAIDIE

Kajang, Selangor | +60 11 5555 2330 | lanzaidie@gmail.com

Portfolio: bit.ly/idlanzafran | **Github:** github.com/IdlanZafran | **LinkedIn:** linkedin.com/in/idlanzafran/

High-achieving Mechanical Engineering undergraduate with specialized expertise in mechanical systems design, predictive analytics, and real-time telemetry. Proficient in ANSYS, MATLAB, and advanced CAD/simulation platforms for computational modeling and structural validation. Proven experience leading multidisciplinary teams to engineer IoT-based SCADA frameworks and integrate complex data pipelines for real-time visualization. Adept at bridging the gap between hardware engineering and data science to optimize system automation, mechatronics, and structural integrity.

EDUCATION

UNIVERSITI KEBANGSAAN MALAYSIA (UKM) | *Expected Graduation: August 2027*

Bachelor of Mechanical Engineering | CGPA: 3.87/4.00 (Current GPA: 3.90/4.00)

TECHNICAL SKILLS

- **Mechanical & FEA:** ANSYS, 3DEXPERIENCE CATIA, Autodesk Inventor, Fusion 360, AutoCAD, Solid Mechanics, Rotating Equipment, CES, Blender (3D Modeling & Simulation).
- **Data Science, Analytics & Databases:** Power BI, R, MATLAB, Google Analytics, Python, Machine Learning (k-NN, K-Means Clustering, Regression).
- **Firmware & Edge Computing:** C/C++, PlatformIO, ESP8266/ESP32, LittleFS, Local HTML/JS Web Servers, Over-The-Air (OTA) Updates.
- **Prototyping & Electronics:** Fritzing, Component-Level Circuit Design, HSE Sensor Integration.

PROFESSIONAL EXPERIENCE

ThingsSentral | UKM | *July 2025 - Present*

Research Assistant & Embedded Systems Developer (Sept 2025 - Present) | Hardware & Firmware Intern (July - Sept 2025)

- **End-to-End Hardware R&D:** Spearheaded the development of custom IoT sensor nodes from scratch, encompassing 3D CAD enclosure design, full-stack firmware programming, and circuit building via Fritzing (integrating capacitors, transistors, fuses, and diodes for stable hardware operation).
- **IoT Fleet Deployment & Power BI Integration:** Engineered an 8-node IoT SCADA architecture to monitor critical infrastructure. Programmed ESP microcontrollers to stream real-time telemetry directly into a Power BI database, establishing a scalable emergency threshold framework applicable to offshore environments.
- **Fail-Safe Edge Computing:** Developed an offline LittleFS data buffering vault, ensuring continuous bulk data logging during network outages and extending backup life to over a week.
- **Enterprise Telemetry & OTA Libraries:** Authored memory-safe C++ telemetry libraries (ThingsSentral, RapidBootWiFi). Published RemoteUpload, a lightweight, cross-platform (ESP32 & ESP8266) library for robust Local and Google Drive-hosted Remote Over-The-Air (OTA) updates.
- **Bi-Directional Telemetry & Access Control:** Engineered a remote command-execution node demonstrating two-way SCADA capabilities. Designed a 12V RFID smart lock with a Telegram-integrated dashboard, optimizing power consumption through a 2-minute Wi-Fi sync cycle.

ENGINEERING PROJECTS

Full-Lifecycle Mechatronics: Autonomous Parcel Sorting Robot | Academic Project | *Sept 2025 - Present*

- **Leadership & Architecture:** Leading a 5-person team to build an autonomous sorting system within a sub-RM750 budget, proposing initial mechanical architecture and taking sole ownership of firmware and system integration.
- **Embedded Systems Programming (C++):** Developed robust firmware for a dual-MCU system, programming the ESP32-CAM to perform real-time QR code scanning and Arduino Mega to execute precise line-following algorithms.
- **Backend Development & Data Integration:** Architected a Python-based backend server using Flask to serve as a central database for sorting operations. Engineered the data pipeline to transmit validated execution payloads and sorting logs directly to a Power BI dashboard for real-time visualization.
- **Autonomous Navigation & AI:** Integrating AI-driven logic with ultrasonic sensors for dynamic, real-time collision avoidance and decision-making.

- **End-to-End Fabrication:** Executed a comprehensive 3-month mechanical design phase (Autodesk Inventor) integrating ANSYS FEA and CES material selection, currently managing Aluminum Alloy hollow frame assembly.

Mechanical Power Transmission & FEA: Double Helical Elevator Gearbox | Academic Project | *May 2025*

- **Design & Validation:** Engineered a double helical transmission system in Inventor, optimizing gear ratios for 98.54% efficiency and a 1.733 m/s lifting speed (50kg load). Validated zero yielding via FEA (safety factor of 4, max Von Mises stress of 10.08 MPa).

Predictive Maintenance: Machine Learning Analysis on Material Stress | Academic Project | *May 2025*

- **ML Modelling:** Developed an R-based linear regression model (R-squared of 0.967) on stress-microstrain data to predict material fatigue. Applied k-Nearest Neighbors and K-Means clustering (Elbow Method) to classify stress risk categories.

Mechanical Design & FEA: Hydro-Powered River Cleanup Conveyor | Academic Project | *Nov 2024 - Jan 2025*

- **Mechanical Design & Force Analysis:** Engineered a conceptual floating, river-current-powered turbine and conveyor system, performing manual static and hydrodynamic calculations to determine critical buoyancy, drag, and mechanical tension.
- **Material Selection & FEA Verification:** Conducted Von Mises stress simulations in Inventor and ANSYS to validate structural integrity. Utilized CES to systematically specify Aluminium Alloy 6061 and generated 2D technical documentation using AutoCAD.

LEADERSHIP & GLOBAL COLLABORATION

International Project Lead | ICAPS (17-University Consortium) | *Jan 2026 - Present*

- **Global Collaboration & Proposal:** Spearheading a multidisciplinary, multinational team (including South Korean counterparts) to conceptualize and pitch an AI-powered manufacturing unit for an ASEAN-wide competition.
- **Mechatronic & AI System Design:** Currently architecting a system utilizing Spatial AI for 3D volume mapping and a PID-controlled thermal press for biodegradable packaging.
- **Project Management:** Managing a USD 700 prototyping budget while coordinating cross-border CAD development and the ongoing physical fabrication of the system.

President | UKM Robocon Club | *Jan 2025 - June 2025*

- **Team Leadership:** Directed mechanical strategies and technical development for a 36-member team, successfully deploying an autonomous robot for the ROBOCON Malaysia 2025 national stage.

AWARDS, CERTIFICATIONS & HIGHLIGHTS

- **Invited Keynote Speaker & Media:** AISCOST Indonesia (Sept 2025). Served as sole Malaysian speaker on "Youth Innovation and Research"; featured in live national TV interviews highlighting technical innovation and robotics.
- **Technical Trainer:** Institut Teknologi Bandung (ITB) (Dec 2025). Led hands-on IoT and SCADA workshops for undergraduate/PhD cohorts.
- **Global Robotics Awards:** Multiple Technical & Emergency Rescue Awards at the International Robotic Olympiad (IRO) in Manila & Chiang Mai.
- **Innovation Competitions:** Top 10 Finalist China-ASEAN Innovation Quest (2025), Gold Medal (6th Sustainability Challenge 2025), 1st Place (MIKEF Gear Up Challenge 2025), Gold Medal GreenCity AR iPad Challenge (2024).
- **Professional Certifications:** Competency Certificate (Level 3) from ThingsSentral Internship Program (Embedded Systems & IoT R&D); Google Analytics for Beginners Certification.
- **Academic Honors & Pipeline:** Academic Honors & Elite Track: Dean's List Award (All Semesters at UKM). Alumnus of Pusat GENIUS@Pintar Negara (selected for the national gifted education program across secondary and pre-university levels; achieved High School Diploma with Dean's List honors).

REFEREES

Prof. Madya Dr. Mohamad Hanif Bin Md Saad
Director, Pusat Teknologi Digital UKM
Universiti Kebangsaan Malaysia
hanifsaad@ukm.edu.my
+60 3 8921 6116

Ir. Dr. Meor Iqram Bin Meor Ahmad
Lecturer, Faculty of Engineering and Built Environment
Universiti Kebangsaan Malaysia
meoriqram@ukm.edu.my
+60 3 8911 8375